



High Availability Clustering

In this lesson, you will cover high availability clustering.

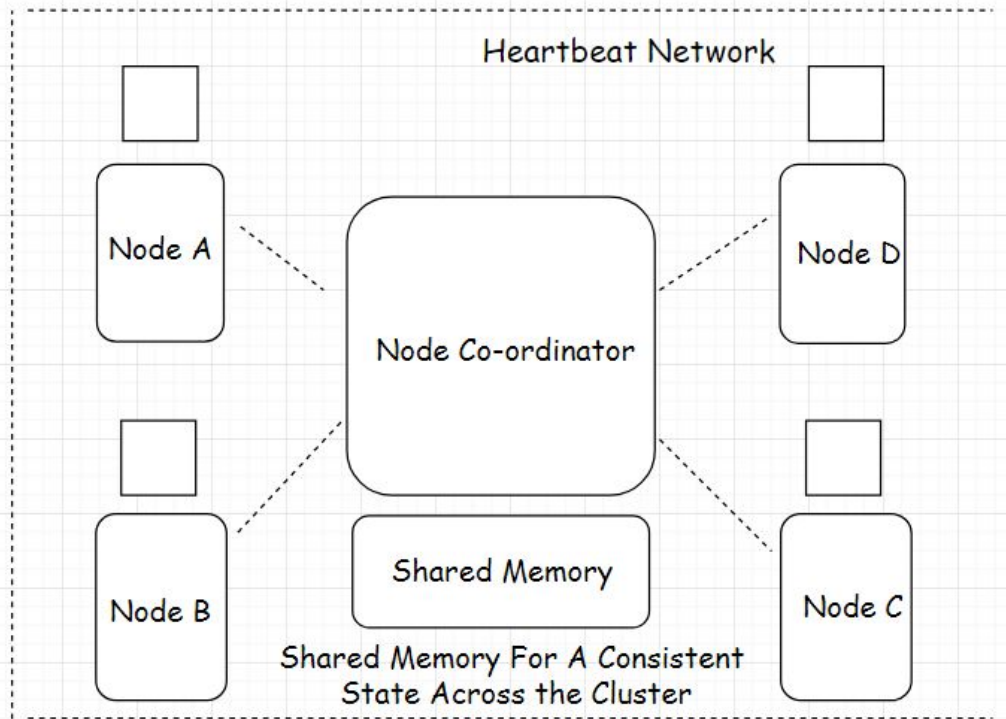
Now that we have a clear understanding of *high availability*, let's talk a bit about the high-availability cluster.

A *high availability* cluster, also known as the *fail-over cluster*, contains a set of nodes running in conjunction with each other that ensures the high availability of the service.

The nodes in the cluster are connected by a private network called the *heartbeat network* that continuously monitors the health and the status of each node in the cluster.

A single state across all the nodes in a cluster is achieved with the help of a shared distributed memory and a distributed coordination service like the *Zookeeper*.





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Highly Available Cluster

To ensure availability, *HA clusters* use several techniques such as *disk mirroring/Redundant Array of Independent Disks (RAID)*, redundant network connections, redundant electrical power, etc. All the possible components having a single point of failure are made redundant to ensure the availability of the service.

Multiple *HA* clusters run together in one geographical zone ensuring minimum downtime and continual service. I've further discussed clustering in great detail in my cloud and systems design course.

Alright, Folks! So now we have a pretty good understanding of *scalability* and

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